



Gary Wu 吳振銓

Senior Full-Stack Engineer · Industrial System Integration Architect

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PROFILE

Senior full-stack engineer with 5+ years specialising in industrial system integration and edge-to-cloud architecture. I integrate edge device data, monitoring workflows, AI inference, and cloud business systems into stable, maintainable, long-term-evolvable digital systems.

HIGHLIGHTS

15+ repositories integrated **3x** query efficiency **↓75%** dev redundancy cut

24h off-site DR cycle **5+** years experience

CORE SKILLS

Frontend	React · Next.js · TypeScript · Redux · D3.js · Tailwind CSS · Electron
Backend & DB	Node.js · Python (FastAPI / Flask) · Strapi · PostgreSQL · MySQL · MongoDB
Industrial & Edge	OPC UA / PLC · SCADA · IPC · Edge Computing · IPFS
AI & Dev	YOLOv8 / v11 · OpenCV · OpenAI API · AI Agent · SDD
Security	ISO 27001:2022 ISMS · Internal Audit · Risk Assessment

EXPERIENCE

Senior Full-Stack Engineer 2020.10 – Present

- Led the solar panel recycling system across 15+ repositories — IPC architecture design and edge-to-cloud cross-system integration
- Connected SCADA, PLC (OPC UA), 5 IPC edge modules, YOLOv11 AI, 3 Cloud APIs, and IPFS
- Introduced config-driven design supporting device versions 1.0–2.x, cutting redundant development by ~75%
- Rebuilt edge logging (levels/categories/events + scheduled cloud upload + SCADA filtering), improving debug localisation efficiency ~3x (~⅔ less time); established off-site cloud DR within 24h RPO

Full-Stack Engineer · LINE Chatbot Developer · IT Lecturer 2020.07 – 2020.12

- Built LINE chatbot automation; served as IT lecturer; participated in full-stack development

Adjunct Assistant

2020.08 - 2020.10

- Assisted with academic research and technical support

SELECTED CASE STUDIES

Solar Panel Recycling System — Edge-to-Cloud Industrial Integration

Spanning 15+ repositories; config-driven design cut dev redundancy ~75%, and log observability made debug localisation ~3x faster.

[Edge-to-Cloud](#) · [IPC](#) · [SCADA](#) · [OPC UA](#) · [Python](#)

Rock System — Full-Stack Business Platform (0→1)

Sole engineer across the SDLC: fine-grained permissions, seat control, 2FA, and containerised delivery.

[React](#) · [Strapi](#) · [PostgreSQL](#) · [Docker](#)

Legacy System Performance Rescue

Located the real bottleneck (client network) and refactored transfer to cut frontend requests to 1/5–1/4.

[Performance](#) · [API Design](#) · [AI Collaboration](#)

Offline iPad Exhibition Setup

Self-hosted a local API on the iPad so video playback became fully independent of venue bandwidth.

[Edge](#) · [Alpine Linux](#) · [Python](#) · [React](#)

CERTIFICATIONS

ISO 27001:2022 ISMS

ISMS implementation, internal audit, risk assessment & treatment.

EDUCATION

National Yang Ming University

Bachelor's degree · Clinical Laboratory Science / Medical Technology

2001 – 2005